

Week 2 Module:

Date: 17/06/2026

1. Find Student Roll Number (Linear Search)

Problem Statement

Your class teacher has a list of student roll numbers stored in the order they registered. Since the list is unsorted, you need to find whether a particular student roll number exists.

Write a program to search for the given roll number using **Linear Search**.

Input Format

- First line contains integer N (number of students)
- Second line contains N roll numbers
- Third line contains integer X (roll number to search)

Output Format

- Print "Found at index i" if present
- Otherwise print "Not Found"

Constraints

- $1 \leq N \leq 1000$

Example

Input

```
5
101 105 113 110 130
110
```

Output

```
Found at index 3
```

2. Search Product ID (Binary Search)

Problem Statement

An online store keeps product IDs sorted in ascending order. A customer wants to know whether a particular product exists in inventory.

Write a program to search the product using **Binary Search**.

Input Format

- First line contains N
- Second line contains sorted product IDs
- Third line contains target ID

Output Format

Print:

Product Available

or

Product Not Available

Constraints

- $1 \leq N \leq 10^5$

Example

Input

```
6
10 20 30 40 50 60
40
```

Output

```
Product Available
```

3. Arrange Exam Scores (Bubble Sort)

Problem Statement

A teacher wants to arrange students' exam scores in ascending order to prepare the result sheet.

Write a program using **Bubble Sort**.

Input Format

- First line contains N
- Second line contains N integers

Output Format

Print the sorted array.

Example

Input

5
89 45 78 12 99

Output

12 45 78 89 99

4. Arrange Library Book Numbers (Insertion Sort)

Problem Statement

A librarian receives book serial numbers one by one and wants to place each new book in the correct position.

Write a program using **Insertion Sort** to arrange book numbers.

Input Format

- First line contains N
- Second line contains N integers

Output Format

Print sorted book numbers.

Example

Input

5
55 23 87 11 34

Output

11 23 34 55 87

5. Rank Players by Score (Selection Sort)

Problem Statement

A gaming club wants to rank players based on scores from lowest to highest.

Write a program using **Selection Sort**.

Input Format

- First line contains N
- Second line contains player scores

Output Format

Print sorted scores.

Example

Input

```
5
70 20 90 10 40
```

Output

```
10 20 40 70 90
```

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① Find student Roll Numbers (Linear Search)

```
#include <stdio.h>
int main () {
    int N, X, found = 0;
    scanf ("%d", &N);
    int roll_numbers[N];
    for (int i = 0; i < N; i++) {
        scanf ("%d", &roll_numbers[i]);
    }
    scanf ("%d", &X);
    for (int i = 0; i < N; i++) {
        if (roll_numbers[i] == X) {
            printf ("found at index %d\n", i);
            found = 1;
            break;
        }
    }
    if (!found) {
        printf ("Not found\n");
    }
    return 0;
}
```

Input :—

5
101 105 113 110 130
110

O/p :— found at index 3

② Search product ID (Binary search)

```
#include <stdio.h>
int main () {
    int N, target, found = 0;
    scanf("%d", &N);
    int product_ids[N];
    for(int i = 0; i < N; i++) {
        scanf("%d", &product_ids[i]);
    }
    scanf("%d", &target);
    int low = 0;
    int high = N - 1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (product_ids[mid] == target) {
            printf("Product Available\n");
            found = 1;
            break;
        }
        else if (product_ids[mid] < target) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    if (!found) {
        printf("Product Not Available\n");
    }
    return 0;
}
```


I/p: — 6

10 20 30 40 50 60
40

O/p: —

product Available.

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④ Arrange Library Book Numbers (Insertion Sort)

```
#include <stdio.h>
int main () {
    int N;
    scanf ("%d", &N);
    int books [N];
    for (int i = 0; i < N; i++) {
        scanf ("%d", &books[i]);
    }
    for (int i = 1; i < N; i++) {
        int key = books[i];
        int j = i - 1;
        while (j >= 0 && books[j] > key) {
            books[j+1] = books[j];
            j--;
        }
        books[j+1] = key;
    }
    for (int i = 0; i < N; i++) {
        printf ("%d", books[i]);
        if (i < N - 1) {
            printf (" ");
        }
    }
    printf ("\n");
    return 0;
}
```

I/p: 5

55 23 87 11 34

O/p: -

11 23 34 55 87

⑤ Rank Players by score (selection sort)

```
#include <stdio.h>
int main () {
    int N;
    scanf("%d", &N);
    int scores[N];
    for (int i=0; i<N; i++) {
        scanf("%d", &scores[i]);
    }
    for (int i=0; i<N-1; i++) {
        int min_idx = i;
        for (int j = i+1; j<N; j++) {
            if (scores[j] < scores[min_idx]) {
                min_idx = j;
            }
        }
        int temp = scores[min_idx];
        scores[min_idx] = scores[i];
        scores[i] = temp;
    }
    for (int i=0; i<N; i++) {
        printf("%d", scores[i]);
        if (i < N-1) {
            printf(" ");
        }
    }
    printf("\n");
    return 0;
}
```

Q/p:-
5

70 20 90 10 40

O/p:-

10 20 40 70 90

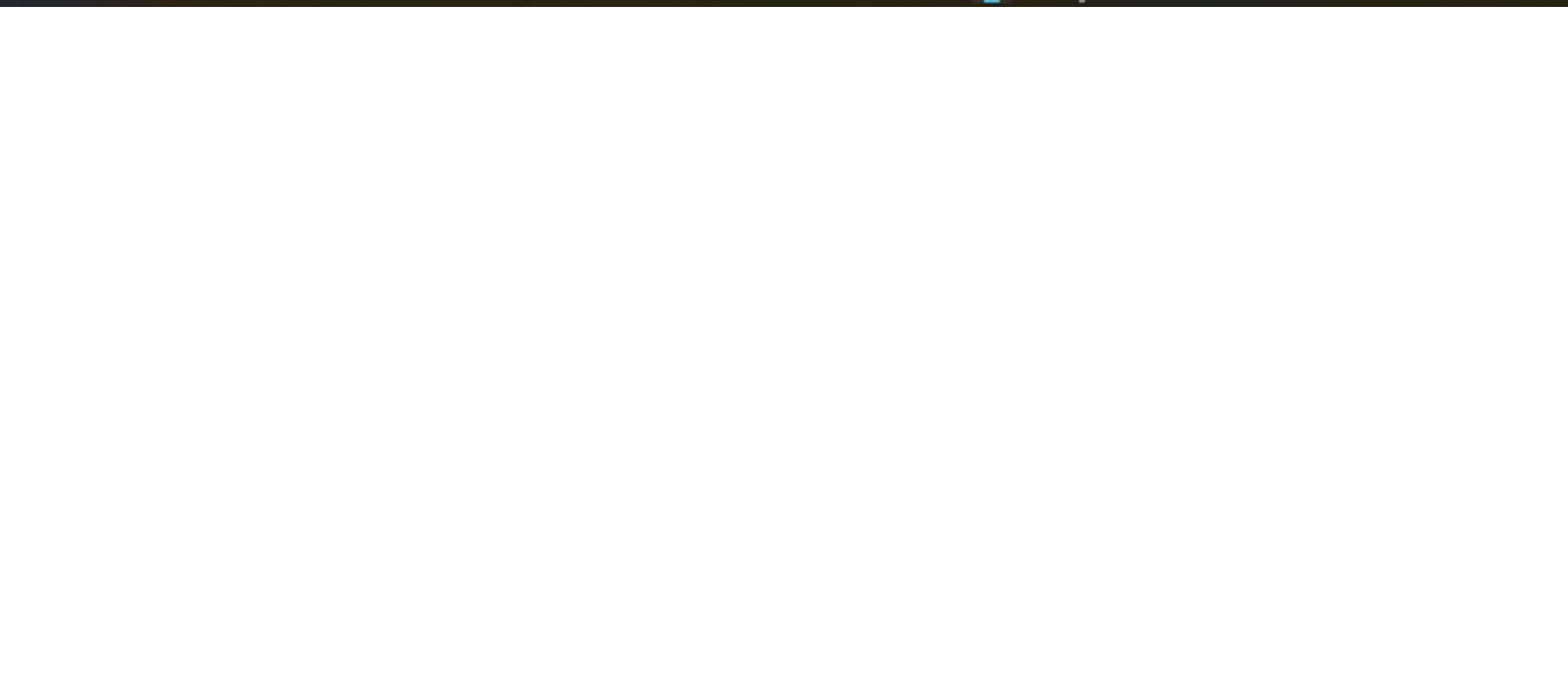
③ Arrange Exam Scores (Bubble Sort)

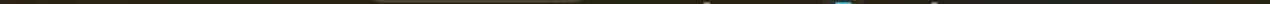
```
#include <stdio.h>
int main () {
    int N;
    scanf ("%d", &N);
    int scores[N];
    for (int i = 0; i < N; i++) {
        scanf ("%d", &scores[i]);
    }
    for (int i = 0; i < N - 1; i++) {
        for (int j = 0; j < N - i - 1; j++) {
            if (scores[j] > scores[j + 1]) {
                int temp = scores[j];
                scores[j] = scores[j + 1];
                scores[j + 1] = temp;
            }
        }
    }
    for (int i = 0; i < N; i++) {
        printf ("%d", scores[i]);
        if (i < N - 1) {
            printf (" ");
        }
    }
    printf ("\n");
    return 0;
}
```

I/p :- 5

89 45 78 12 99

O/p :- 12 45 78 89 99





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Browser address bar: https://www.onlinegdb.com/online_c_compiler#

Browser toolbar: Run, Debug, Stop, Share, Save, Beautify, Language C

Code editor: main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int scores[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &scores[i]);
10    }
11
12    for (int i = 0; i < N - 1; i++) {
13        for (int j = 0; j < N - i - 1; j++) {
14            if (scores[j] > scores[j + 1]) {
15                int temp = scores[j];
16                scores[j] = scores[j + 1];
17                scores[j + 1] = temp;
18            }
19        }
20    }
21
22    for (int i = 0; i < N; i++) {
23        printf("%d", scores[i]);
24        if (i < N - 1) {
25            printf(" ");
26        }
27    }
28    printf("\n");
29
30    return 0;
31 }
```

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main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int books[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &books[i]);
10    }
11
12    for (int i = 1; i < N; i++) {
13        int key = books[i];
14        int j = i - 1;
15
16        while (j >= 0 && books[j] > key) {
17            books[j + 1] = books[j];
18            j--;
19        }
20        books[j + 1] = key;
21    }
22
23    for (int i = 0; i < N; i++) {
24        printf("%d", books[i]);
25        if (i < N - 1) {
26            printf(" ");
27        }
28    }
29    printf("\n");
30
31    return 0;
32 }
```

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Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int books[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &books[i]);
10    }
11
12    for (int i = 1; i < N; i++) {
13        int key = books[i];
14        int j = i - 1;
15    }
```

input

```
5
55 23 87 11 34
11 23 34 55 87

...Program finished with exit code 0
Press ENTER to exit console.
```

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Run Debug Stop Share Save Beauty

main.c Language C

```
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int scores[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &scores[i]);
10    }
11
12    for (int i = 0; i < N - 1; i++) {
13        int min_idx = i;
14        for (int j = i + 1; j < N; j++) {
15            if (scores[j] < scores[min_idx]) {
16                min_idx = j;
17            }
18        }
19        int temp = scores[min_idx];
20        scores[min_idx] = scores[i];
21        scores[i] = temp;
22    }
23
24    for (int i = 0; i < N; i++) {
25        printf("%d", scores[i]);
26        if (i < N - 1) {
27            printf(" ");
28        }
29    }
30    printf("\n");
31
32    return 0;
33 }
```

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Toolbar: Run, Debug, Stop, Share, Save, Beautify

Language: C

```
main.c
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int scores[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &scores[i]);
10    }
11
12    for (int i = 0; i < N - 1; i++) {
13        int min_idx = i;
14        for (int j = i + 1; j < N; j++) {
15            if (scores[j] < scores[min_idx]) {
16                min_idx = j;
17            }
18        }
19    }
20 }
```

input

```
5
70 20 90 10 40
10 20 40 70 90
```

...Program finished with exit code 0
Press ENTER to exit console.

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